README

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Open-Source Nematode DrugBase (ONDB) — Full Pipeline Documentation

Overview

The Open-Source Nematode DrugBase (ONDB) is a fully reproducible, contract-driven data pipeline designed to integrate genomic, functional, orthology, and phenotypic information across parasitic nematodes. The system unifies **BioMart-derived structural and orthology data** with **WormMine-derived RNAi phenotypic evidence** from *Caenorhabditis elegans* to support target prioritisation for neglected tropical disease drug discovery.

This README documents the full architecture as of **23 November 2025**, including new modules, data flow, contracts, provenance, and the integration strategy for C. elegans RNAi phenotypes.

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1. Architectural Structure

The pipeline is divided into three major subsystems:

1. BioMart modules (structural + orthology + protein domains)

These fetch all genomic and functional annotations required for each parasite species:

- Principal gene list
- Coding sequence (CDS) boundaries
- Human orthologues
- *C. elegans* orthologues
- InterPro domains
- InterPro→GO mapping
- GO DAG propagation

2. WormMine modules (functional phenotype extraction)

This subsystem retrieves RNAi phenotypes from curated *C. elegans* RNAi screens via the WormMine REST interface:

- Uses template-based XML queries identical to the WormBase WormMine QueryBuilder
- Produces long-format RNAi tables
- Aggregates phenotypes into per-gene flags (e.g., lethal, sterile, locomotion)

3. Integration subsystem (run_v01.py)

This orchestrates:

- Species-wise execution of all BioMart and WormMine modules
- Validation and deterministic merging
- Per-gene consolidation
- Final export table
- Provenance metadata (run folders, URLs, version numbers, logs)

The following diagram summarises the data flow:

```
            +------------------+
            | Species list    |
            +------------------+
                   |
          --------------------------------
          |                    |
     +-------------+        +----------------+
     |  BioMart  |          |  WormMine    |
```

```
|  (per-spec) |              | (C. elegans)   |
+-------------+              +----------------+
| Gene list   |              | RNAi XML       |
| CDS         |              | REST query     |
| Human orth. |              | Long-format    |
| CE orth.    |              | Aggregation    |
| InterPro    |              +----------------+
| GO mapping  |                      |
+-------------+                      |
      |                              |
      ------------ merged ---------------
                 |
         +---------------------+
         |    Final export     |
         |    (per species)    |
         +---------------------+
```

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2. Core Development Contract (ONDB Pipeline Contract)

All ONDB modules must satisfy the following rules:

2.1 BioMart XML consistency
- XML templates must be used exactly as provided by BioMart.
- The only allowed modifications:
  1. `header="1"` (mandatory)
  2. Insert `{species_code}` into species filter attributes
- Do NOT modify any attribute order, whitespace, or structure.

2.2 Column discovery and validation
- Expected column names must come **from the XML template itself**, not hard-coded.
- After fetching, the module must:
  - Print the header line
  - Print a numbered list of column names
  - Validate required columns with alias-based drift correction

2.3 Deterministic row selection
- One row per gene after merging.
- Ties resolved deterministically (highest % identity, then lexical fallback).

2.4 Single-species input
- Species codes must be supplied exactly one at a time.
- Comma-separated species codes are rejected.

2.5 Polars internal → Pandas boundary
- All internal processing uses **Polars** for performance.
- Each module returns **Pandas** to the boundary for merging.

2.6 Provenance
- Downloaded auxiliary files (InterPro→GO, GO DAG) are stored in the run folder.
- Versions and URLs logged into `run.json`.

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3. BioMart Modules

Below is the updated full documentation matching today’s pipeline state.

3.1 Principal Gene List

Purpose
Retrieve major structural gene information per species.

Source
BioMart: `wbps_gene` dataset.

XML Attributes
- `Gene stable ID`
- `Transcript stable ID`
- `Transcript biotype`
- `Gene description`
- `Genome name`

Validation
- Required columns discovered at runtime.
- Only protein-coding transcripts retained.
- Principal isoforms selected via deterministic rule:
  - Longest CDS
  - If equal, pick lowest transcript ID lexically

Output
One row per parasite gene.

3.2 Coding Sequences (CDS)

Purpose
Retrieve CDS boundaries for defining principal isoforms.

Attributes
- `CDS start (within cDNA)`
- `CDS end (within cDNA)`

Rules
- Multi-exonic CDS reconstructed from Polars groupby.
- Deterministic isoform selection.

3.3 Human Orthologues

XML attributes
- `Human gene stable ID`
- `Human gene name`
- `Homology type`
- `% identity`
- `Human % identity`

Rules

- Aliased columns allowed (`Human % identity` sometimes becomes `% identity r1`).

- One row per parasite gene post-resolution.

3.4 C. elegans Orthologues

XML attributes

- `Caenorhabditis elegans (PRJNA13758) [WS290] gene stable ID`

- `gene name`

- `Homology type`

- `% identity`

- CE-specific % identity

Output

Added to export as:

- `ce_gene_id`

- `ce_gene_name`

- `best_ce_identity_homology_type`

- Flags:

  - `lacks_celegans_orthologue`

  - `best_celegans_orthologue_lt_40pct_identity`

3.5 InterPro Domains

XML attributes

- `InterPro ID`

- `InterPro description`

Multiple rows per gene allowed.

3.6 GO Annotations

Inputs

- `interpro2go` mapping (IPR → GO)

- GO DAG (is_a hierarchy)

Outputs

Columns with boolean flags:

- `is_enzyme`

- `is_kinase`

- `is_gpcr`

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4. WormMine Modules (NEW, 23 Nov 2025)

4.1 Motivation

C. elegans provides the most comprehensive genome-wide RNAi phenotyping available. WormMine exposes these data via a declarative XML query format. Integrating these phenotypes allows ONDB to prioritise parasite genes supported by functional evidence.

Functional questions supported:
- Is the orthologue essential?
- Does knockdown cause sterility, embryonic lethality, locomotion defects?
- Are phenotypes consistent across multiple RNAi screens?

4.2 XML Template for WormMine (canonical)

ONDB uses *the exact XML query downloaded from WormMine*:

```xml
<query model="genomic" view="
  Gene.primaryIdentifier
  Gene.secondaryIdentifier
  Gene.RNAiResult.primaryIdentifier
  Gene.RNAiResult.phenotype.identifier
  Gene.RNAiResult.phenotype.name
  Gene.RNAiResult.phenotypeRemark
  Gene.RNAiResult.remark"
  sortOrder="Gene.primaryIdentifier ASC">
 <constraint path="Gene.primaryIdentifier" op="=" value="WBGene00006757" code="A" />
</query>
```

Only the value of `Gene.primaryIdentifier` is changed programmatically.

4.3 WormMine REST API (Python 3.12–compatible)

Because the official `intermine` Python package is Python 2–oriented and incompatible with Python 3.12 (`urlparse`, `MutableMapping` deprecations), ONDB uses a **custom REST client**:

- Sends XML payload to:
 ```
 http://intermine.wormbase.org/tools/wormmine/service/query/results?format=json
 ```
- Robust against:
 - HTTP 500 errors
 - Timeouts
 - Restart of server connection
 - Empty results

4.4 Long-Format Results

Each RNAi result is returned as one row:

Columns:
- `wormbase_gene_id`
- `sequence_name`
- `rnai_result_id`
- `phenotype_id`

- `phenotype_name`
- `phenotype_remark`
- `rnai_result_remark`

Multiple rows per CE gene are expected.

4.5 Batch Parallel Strategy C (Default for ONDB)

Motivation
WormMine handles many small requests better than one huge request.

Strategy
1. Deduplicate CE gene IDs
2. Break into batches of size 20
3. Sequential batches, threads inside each batch
4. Truncated exponential backoff for retries

4.6 Aggregation per C. elegans Gene

The long table is collapsed to one row per CE gene.

Phenotypic flags generated:
- `ce_rnai_any_lethal`
- `ce_rnai_any_emb` (embryonic lethal)
- `ce_rnai_any_sterile`
- `ce_rnai_any_locomotion`

Rules:
- A phenotype is considered present if **any** RNAi experiment mapped to that category.

Unused phenotype names are kept in a parallel list column for future search functionality.

4.7 Missing CE orthologues

If a parasite gene lacks valid CE orthologues:
- No WormMine call is made
- All CE RNAi flags default to `False`

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5. Integration Subsystem (run_v01.py)

Full process for each species

5.1 Stepwise workflow
1. Load species list
2. For each species:
   - Gene list
   - CDS
   - Human orthologues

  - CE orthologues
  - InterPro / GO
  - Merge all
  - CE RNAi selection:
    - Extract CE gene IDs
    - Fetch RNAi data
    - Aggregate
  - Attach phenotype flags
3. Concatenate all species
4. Export final table + run folder metadata

5.2 Error Handling
- Column drift warnings
- BioMart unavailable
- XML mismatch
- WormMine 500 errors
- Timeout → retry
- Missing orthologues

5.3 Provenance Logging
Each run folder contains:
- `run.json`
- raw BioMart XML queries
- exact URLs
- InterPro→GO copies
- GO DAG copy
- outputs per species
- combined export

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6. Final Export Format

Columns (example, 42 columns)
- Structural columns
- Human orthology columns
- CE orthology columns
- CE RNAi phenotype flags
- Functional domain flags

Each row = one parasite gene.

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7. Known Limits and Future Work
- WormMine rate limiting still variable
- Phenotype grouping: future expansion to full ontology

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8. Citation and Credits

The ONDB pipeline is designed for transparent and reproducible target discovery, integrating open data from:

- WormBase ParaSite BioMart
- WormBase WormMine
- Gene Ontology
- InterPro
- ZINC/ChEMBL future integration